

CAITLYN'S COFFEE WAREHOUSE

End-to-End Business Intelligence Project

Raw Excel data -> Power Query -> Data Model -> DAX -> Interactive Power BI reporting

POWER BI

POWER QUERY

DAX

DATA MODELING

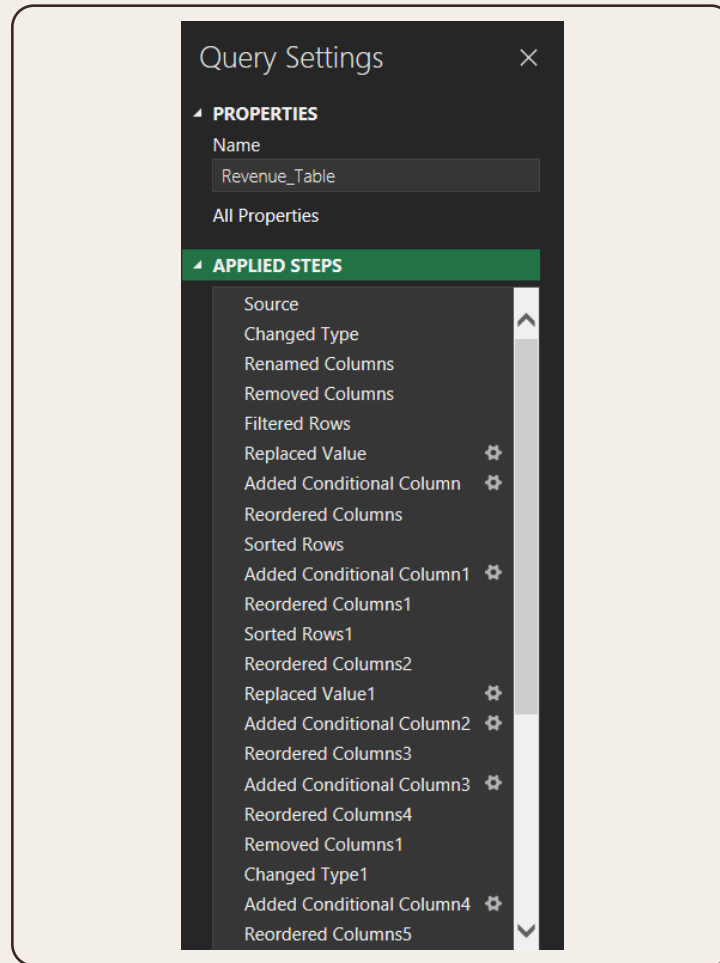
EXCEL

Executive Overview | Invoice & Expense Operations | Sales & Customer Performance

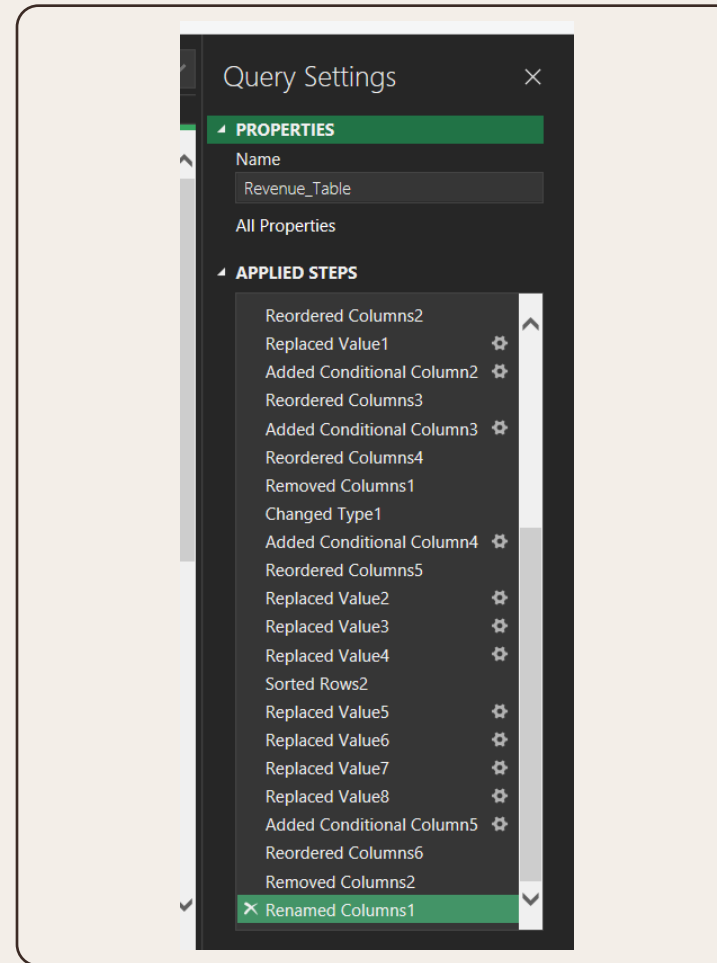
POWER QUERY TRANSFORMATION

FULL TRANSFORMATION HISTORIES FROM THE REVENUE AND COST QUERIES

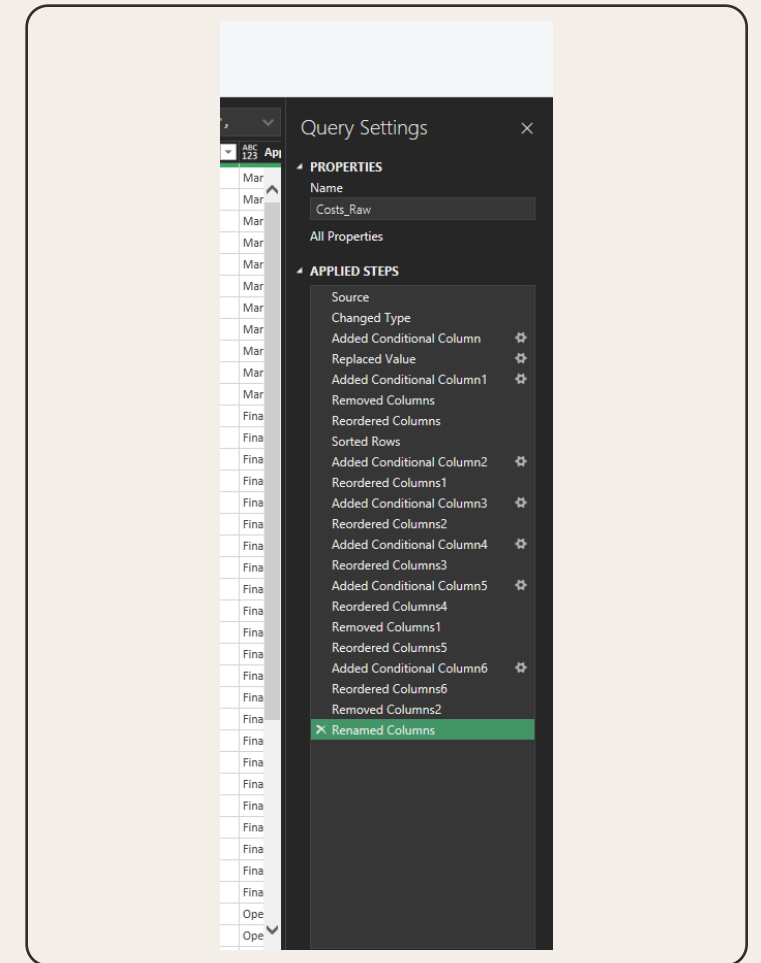
The full Applied Steps panels show the depth and repeatability of the cleaning workflow.



Revenue query - upper steps



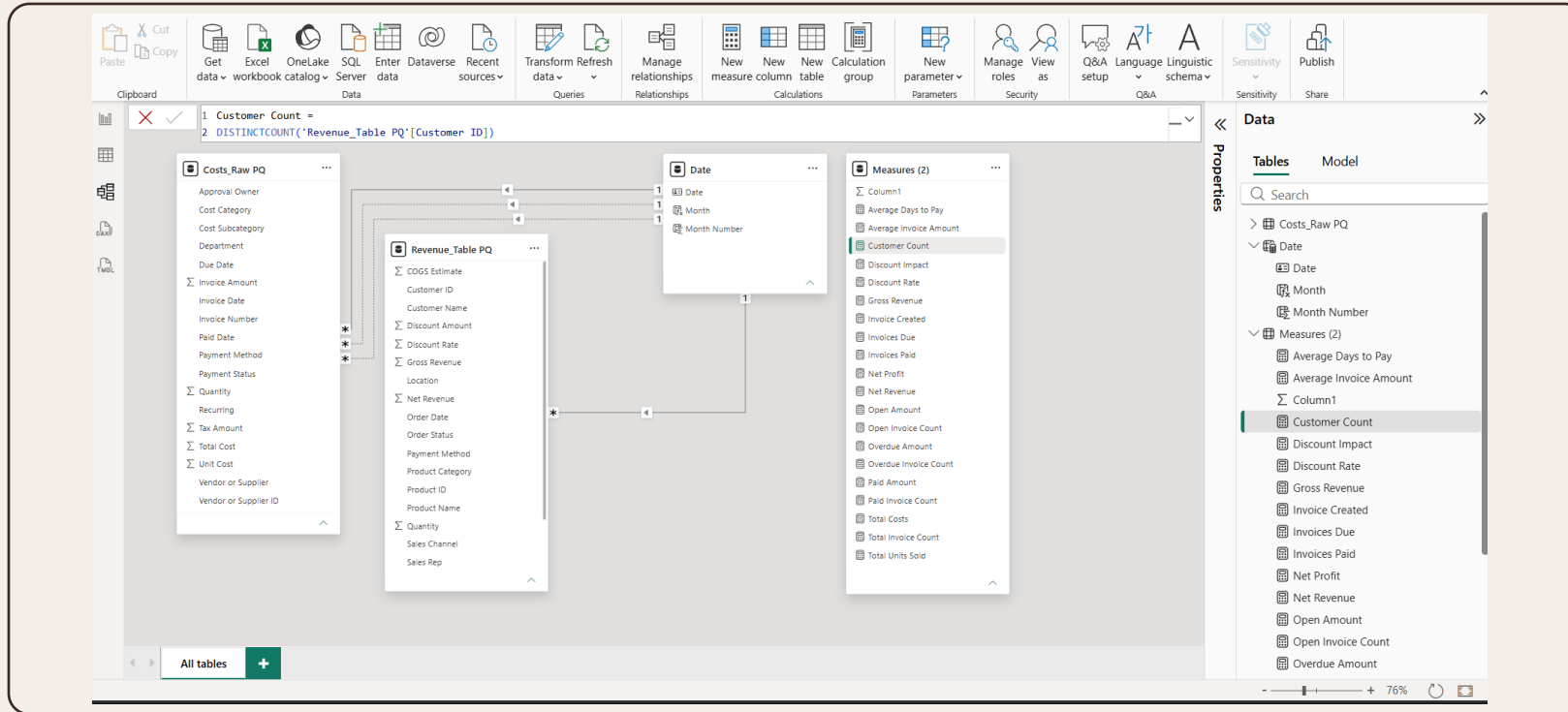
Revenue query - lower steps



Costs query - full steps

DATA MODEL + MEASURES

FULL POWER BI MODEL VIEWS AND REUSABLE KPI LOGIC



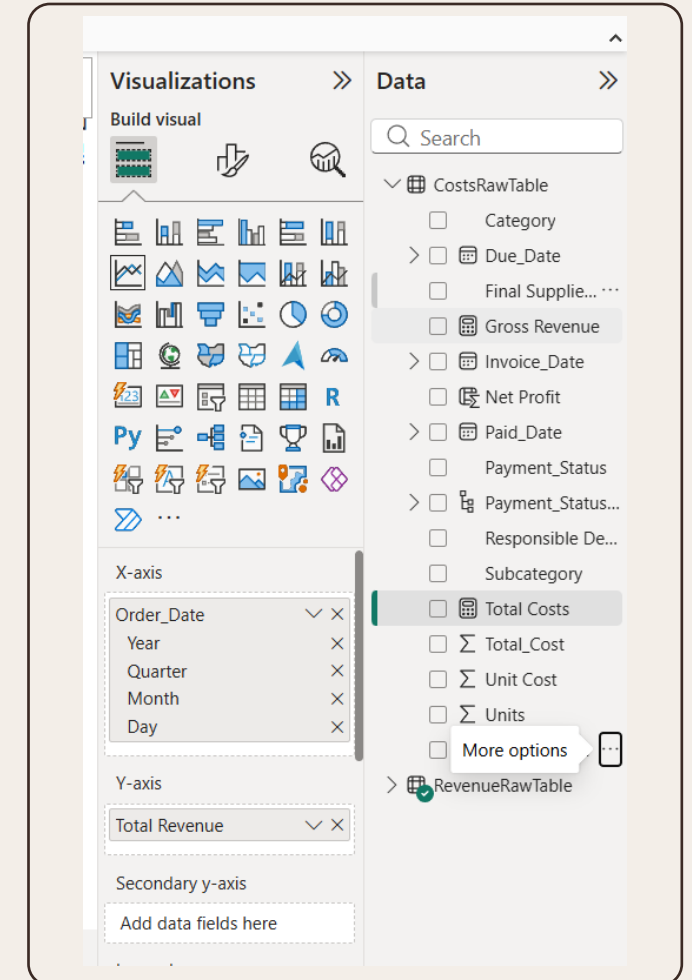
Modeling approach

Date dimension supports consistent monthly analysis.

Revenue and cost tables are related to a shared date structure.

Dedicated Measures table centralizes KPI and operational calculations.

Inactive relationships support Due Date and Paid Date lifecycle analysis.



DAX BUSINESS LOGIC

EXAMPLE MEASURE: AVERAGE DAYS TO PAY

The screenshot displays the Microsoft Power BI Desktop interface. The top ribbon shows the 'Home' tab with various data and calculation tools. The main workspace is divided into three panes: the left pane shows the 'Model' view with a table named 'Revenue_Table PQ' and its columns; the center pane shows the DAX formula editor for the 'Average Days to Pay' measure; the right pane shows the 'Properties' and 'Data' panes.

DAX Formula:

```
1 Average Days to Pay =  
2 AVERAGEX(  
3     FILTER(  
4         'Costs_Raw PQ',  
5         'Costs_Raw PQ'[Payment Status]= "Paid"  
6         && NOT ISBLANK('Costs_Raw PQ'[Paid Date])  
7     ),  
8     DATEDIFF(  
9         'Costs_Raw PQ'[Invoice Date],  
10        'Costs_Raw PQ'[Paid Date],  
11        DAY  
12    )  
13 )
```

Properties Pane:

- Name:** Average Days to Pay
- Home table:** Measures (2)
- Description:** Enter a description
- Synonyms:** average days to pay, average days
- Display folder:** Enter the display folder
- Is hidden:** (checkbox)

Data Pane:

- Tables:** Costs_Raw PQ, Date
- Measures (2):** Average Days to Pay, Average Invoice Amount, Column1, Discount Impact, Discount Rate, Gross Revenue, Net Profit, Net Revenue, Open Amount, Open Invoice Count, Overdue Amount, Overdue Invoice Count, Paid Amount, Paid Invoice Count, Total Costs, Total Invoice Count
- Revenue_Table PQ:** COGS Estimate, Customer ID, Customer Name, Discount Amount, Discount Rate, Gross Revenue, Location, Net Revenue, Order Date

Why this matters

The measure filters to invoices marked Paid, excludes blank Paid Dates, and calculates the average elapsed days from Invoice Date to Paid Date using AVERAGEX and DATEDIFF.

EXECUTIVE OVERVIEW

LEADERSHIP VIEW OF FINANCIAL HEALTH AND PERFORMANCE

CAITLYN'S COFFEE WAREHOUSE EXECUTIVE OVERVIEW

\$494.0K

Net Revenue

\$288.5K

Total Costs

\$205.5K

Net Profit

\$6.7K

Overdue Amount

5

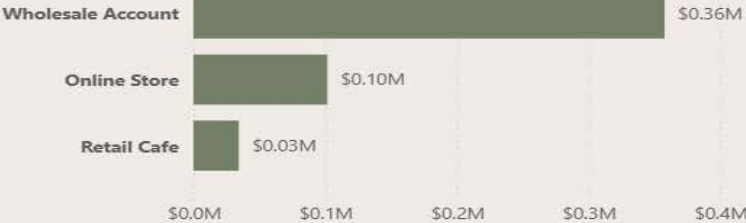
Open Invoice Count

Monthly Revenue vs. Costs

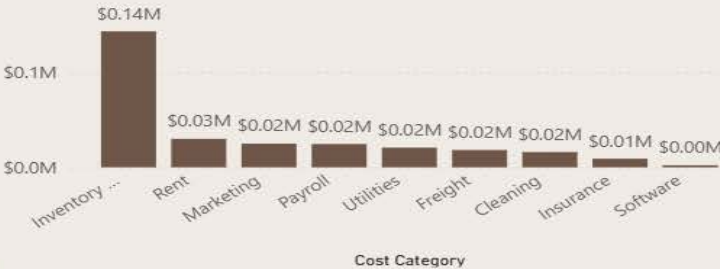
● Net Revenue ● Total Costs



Net Revenue vs. Sales Channel



Total Costs by Category



Location

Charlottesville Warehouse

Online

Farmers Market

Richmond Pop-Up

Month

All

\$494K Net Revenue

\$288.5K Total Costs

\$205.5K Net Profit

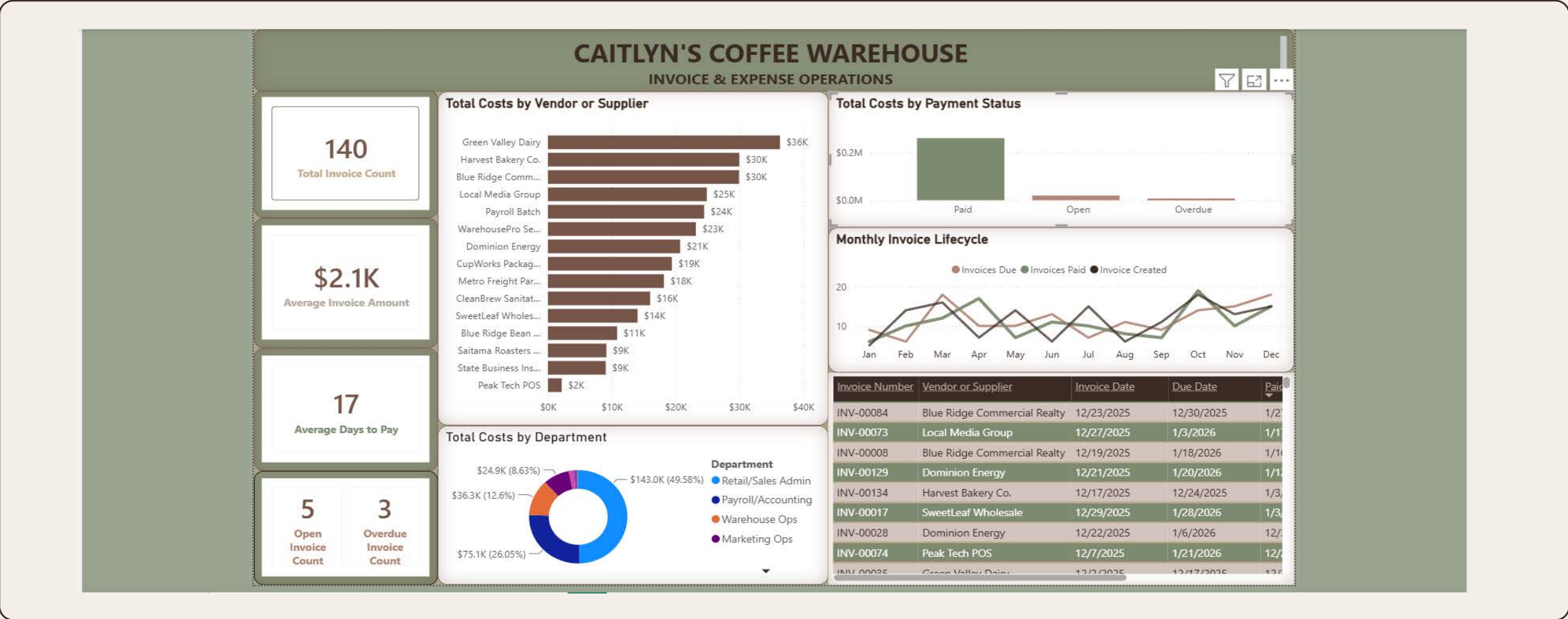
\$6.7K Overdue

5 Open Invoices

Monthly trends, channel performance, cost categories and interactive location/month filtering.

INVOICE & EXPENSE OPERATIONS

PAYMENT TIMING, VENDOR SPEND AND INVOICE-LEVEL CONTROL



140 Invoices

\$2.1K Avg. Invoice

17 Avg. Days to Pay

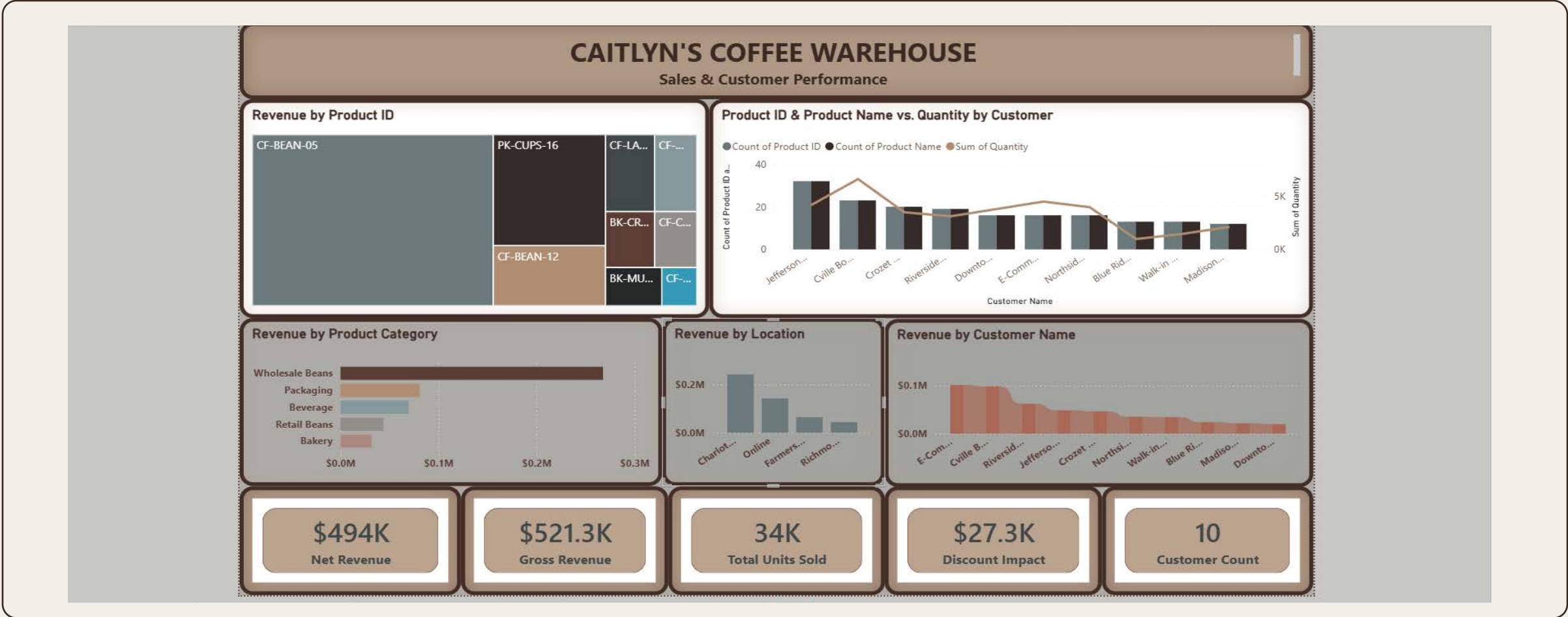
5 Open

3 Overdue

Vendor spend, payment status, departmental costs, invoice lifecycle and transaction-level detail.

SALES & CUSTOMER PERFORMANCE

CUSTOMER, PRODUCT, LOCATION AND DISCOUNT ANALYSIS



END-TO-END ANALYTICS WORKFLOW

Source -> Clean -> Model -> Calculate -> Visualize -> Validate

Skills demonstrated

Power BI

Power Query

DAX

Excel

Data Cleaning

Data Transformation

Data Modeling

KPI Development

Business Intelligence

Data Visualization

Analytical QA

Business Analysis

Built as a simulated wholesale coffee business case study. The project demonstrates how raw operational and sales data can be transformed into a validated, interactive reporting solution for executive, expense-management and customer-sales decision support.